

A comparison of the effect of diets containing beef protein and plant proteins on blood lipids of healthy young men.

The effect of plant and animal protein on blood lipid levels was investigated in eight healthy normolipidemic men aged 18 to 27 yr. All subjects were fed both plant and animal protein diets in a cross-over design. Each diet was consumed for a 21-day period. Proteins from commonly used plant sources made up the plant protein diet. Beef protein was substituted for 55% of the plant proteins in the animal protein diet. Fasting venous blood samples were collected at the beginning of the study and at 7-day intervals throughout the 42-day study. Serum was analyzed for total cholesterol and triglycerides. Plasma low-density and high-density lipoprotein cholesterol were determined. There were not any statistically significant differences in mean serum total cholesterol or mean plasma low-density lipoprotein cholesterol when subjects consumed the diets. Mean plasma high-density lipoprotein cholesterol levels were significantly (p less than 0.05) elevated at the end of the 21-day period when the animal protein diet was consumed (48 ± 3 mg/dl) compared to the period when the plant protein diet was fed (42 ± 2 mg/dl). Mean serum triglyceride values were significantly (p less than 0.05) increased at day 7 of the plant protein diet period (136 ± 19 mg/dl) compared to the same time period when the animal protein diet was consumed (84 ± 12 mg/dl). The results of the study indicated that the ingestion of a diet in which 55% of the protein was supplied by beef protein was not associated with a hypercholesterolemic effect in healthy normolipidemic young men.